

I think the experiment's motivation is handling the inequality of city activity during the weekdays: the higher activity of Gotham weeknights can be better served if driver partners in Metropolis had fewer barriers to Gotham, and the inverse could be said of Metropolis weekdays.

- 1) Tracking intercity service means tracking the number of trips from one city to another. Comparing this number when toll costs are or aren't reimbursed provides an excellent proxy for whether reimbursing toll costs encourages driver partners to serve both cities.
- 2) Designing a practical experiment
 - a) One possible design has half of Gotham driver partners and half of Metropolis driver partners randomly selected and notified that their toll costs would be reimbursed. However, two issues: fairness, and Ultimate must eventually choose whether to reimburse toll costs for all driver partners (a different situation from the half-and-half arrangement with all the confounding variables that entails). To avoid these issues, **I would instead reimburse all toll costs for all driver partners for one year and compare the number of intercity trips with that of the previous year.** There remains the problem that one year isn't the same as the next (and all the confounding variables that entails), but there are certainly fewer ethical issues about treating driver partners differently based on chance.
 - b) We can't make normality assumptions, so bootstrapping is best. The null hypothesis would be that the number of intercity trips would not differ whether reimbursing all toll costs for all driver partners.
 - c) Caveats include profits: does reimbursing toll costs provide greater returns? If so, no problem. If not, under what conditions would Ultimate step back reimbursements? For example, it's conceivable that the losses from reimbursing toll costs would be short-term.